Planning a Software Project



Software Project

- Goal: Build a software system to meet commitments on cost(budget), schedule (time), quality
- Worldwide - many projects fail
  - one-third are runaways with cost or schedule overrun of more than 125%

Project Failures

- Major reasons for project runaways
  - unclear objectives
  - bad planning
  - no project management methodology
  - new technology
  - insufficient staff
- All of these relate to project management
- Effective project management is key to successfully executing a project

Why improve PM?

- Better predictability leading to commitments that can be met
- Lower cost through reduced rework, better resource mgmt, better planning,..
- Improved quality through proper quality planning and control
- Better control through tracking changes , monitoring progress etc.

Why improve PM ....

- Better visibility into project health and state, leading to timely intervention
- Better handling of risks reducing the chances of failure
- All this leads to higher customer satisfaction
- And organization improvement

The Project Mgmt Process

- Has three phases –
  - planning,
  - monitoring and control,
  - closure
- Planning is done before most of the engineering process (life cycle, LC) and closure after the process
- Monitoring phase is in parallel with LC
- We focus on planning; monitoring covered through its planning

Project Planning

- Basic objective: To create a plan to meet the commitments of the project, I.e. create a path that, if followed, will lead to a successful project
- Planning involves defining the LC process to be followed, estimates, detailed schedule, plan for quality, etc.
- Main output - a project management plan and the project schedule

Key Planning Tasks

- Estimate effort
- Define project milestones(checkpoint goals) and create a schedule
- Define quality objectives and a quality plan
- Identify risks and make plans to mitigate them(reduce)
- Define measurement plan, project-tracking procedures, training plan, team organization, etc.

Effort Estimation



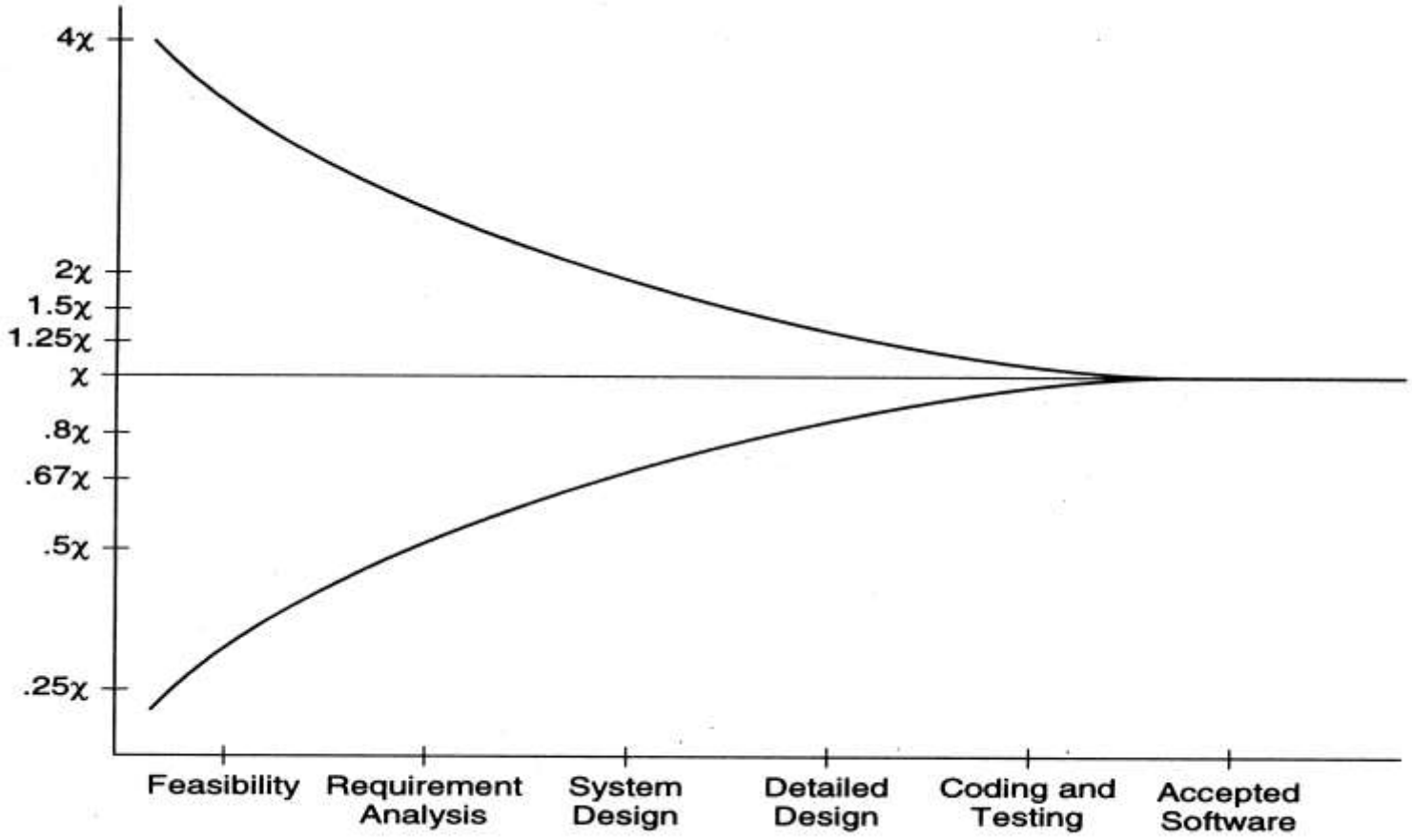
Effort Estimation

- For a project, total cost and duration has to be committed in start
- Requires effort estimation, often in terms of person-months
- Effort estimate is key to planning - schedule, cost, resources depend on it
- Many problems in project execution stem from improper estimation

Estimation..

- No easy way, no silver bullet
- Estimation accuracy can improve with more information about the project
- Early estimates are more likely to be inaccurate than later
  - More uncertainties in the start
  - With more info, estimation becomes easier

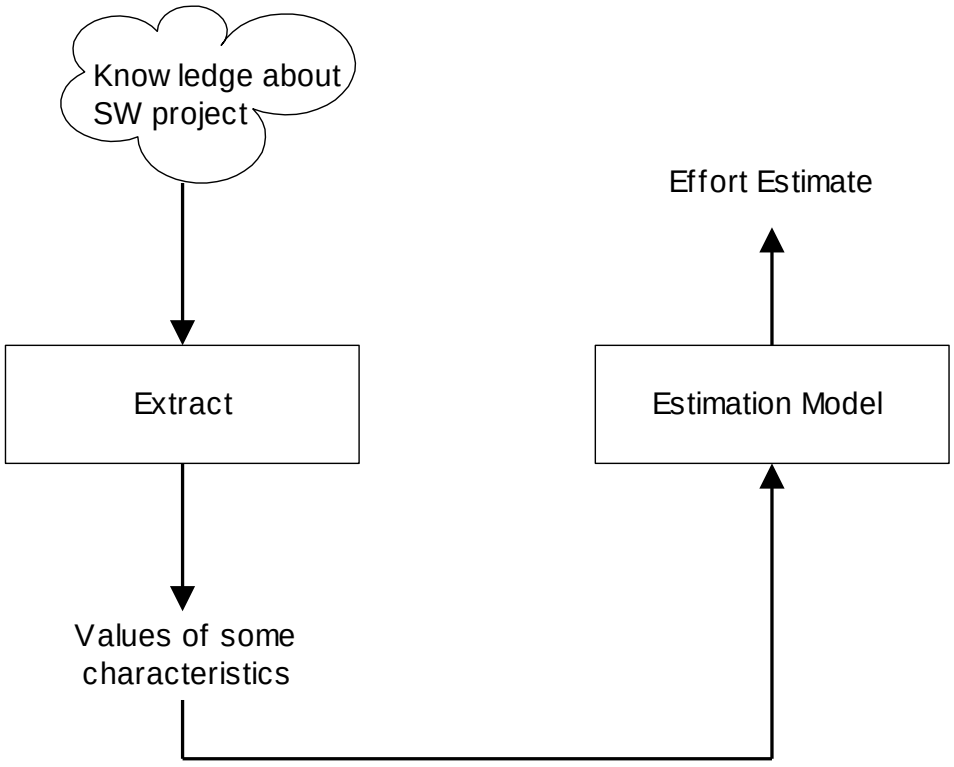
Estimation accuracy



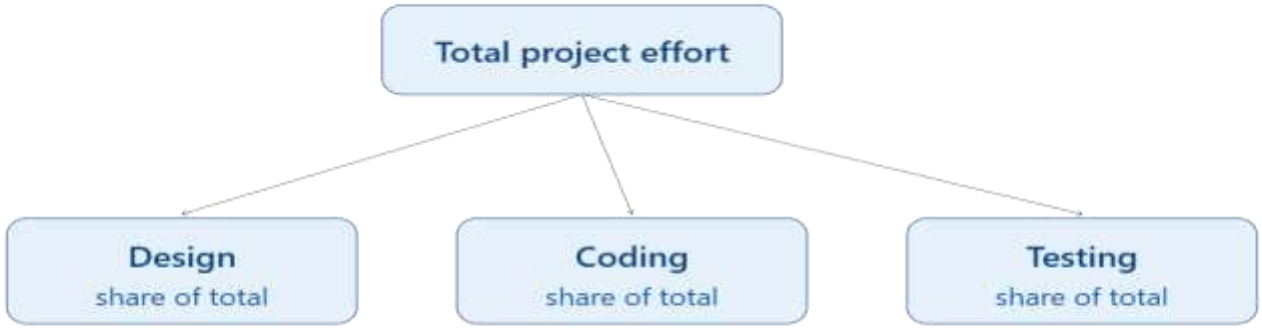
Effort Estimation Models..

- A model tries to determine the effort estimate from some parameter values
- A model also requires real input about the project, and cannot work in vacuum
- So to apply a model, we should be able to extract properties about the system
- Two types of models - top-down and bottom-up

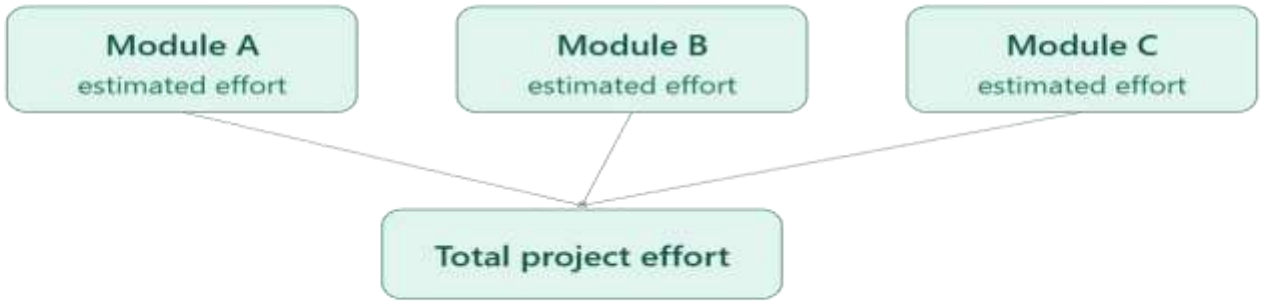
Effort Estimation Models



Top-down (e.g. COCOMO)



Bottom-up (e.g. decomposition-based)



Top down estimation

- First determines the total effort, then effort for components
- Simple approach – estimate effort from size and productivity
  - Get the estimate of the total size of the software
  - Estimate project productivity using past data and project characteristics
  - Obtain the overall effort estimate from productivity and size estimates
- Effort distribution data from similar project are used to estimate effort for different phases

Top-down Estimation

- A better method is to have effort estimate as a function of size using:

$$\text{Effort} = a * \text{size}^{b}$$

- E is in person-months, size in KLOC
- Incorporates the observation that productivity can dip with increased size
- Constants a and b determined through regression analysis of past project data

Bottom-up Estimation

- An alternate approach to top-down
- Effort for components and phases first estimated, then the total
- Can use activity based costing - all activities enumerated and then each activity estimated separately
- Can group activities into classes - their effort estimate from past data

An Estimation Procedure

- Identify programs in the system and classify them as simple, medium, or complex (S/M/C)
- Determine the average coding effort for S/M/C
- Get the total coding effort.
- Use the effort distribution in similar projects to estimate effort for other tasks(design, testing) and the total effort.
- Refine the estimates based on project-specific factors

Scheduling and Staffing

Project Schedule

- **Describes** the software-development cycle for a particular project by

1) **enumerating** the **phases or stages** of the project.
2) **breaking each phase** into **discrete tasks or activities** to be completed.

- **Portrays** the **interactions** among the **activities** and **estimates** the **times** that each task or activity will take.

Project Schedule: Approach

- **Understanding customer's needs** by **listing** all **project deliverables**

1) Documents
2) Demonstrations of function
3) Demonstrations of subsystems
4) Demonstrations of accuracy
5) Demonstrations of reliability, performance or security

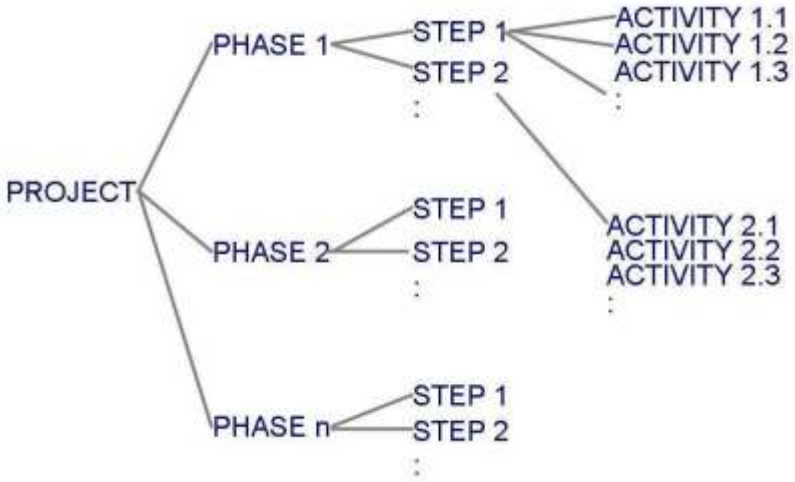
- **Determining milestones** and **activities** to produce the deliverables.

Milestones and activities

- **Activity**: takes place over a period of time.
- **Milestone**: completion of an activity -- a particular point in time.
- **Precursor**: **event or set of events** that must occur in order for an **activity to start**.
- **Duration**: length of time needed to complete an activity.
- **Due date**: date by which an activity must be completed.

Project Schedule

- **Project** development can be separated into a succession of **phases** which are composed of **steps,** which are composed of **activities.**



Critical Path Method (CPM)

- Minimum amount of time it will take to complete a project
  - Reveals those activities that are most critical to completing the project on time.
- Real time (actual time): estimated amount of time required for the activity to be completed.
- Available time: amount of time available in the schedule for the activity's completion
- Slack time: the difference between the available time and the real time for that activity.

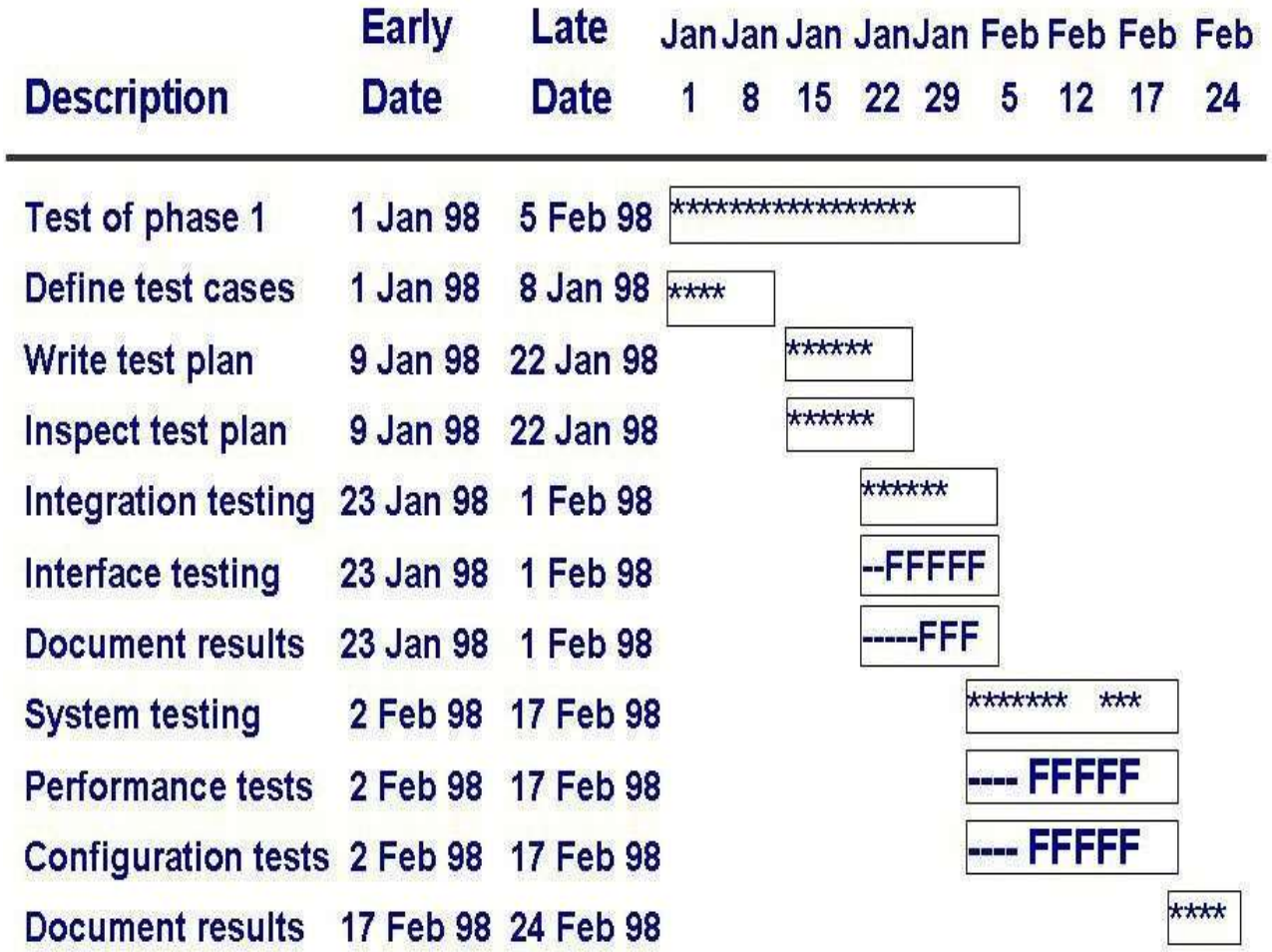
Critical Path Method (CPM)

- **Critical path**: the slack at every node is zero
  – can be more than one in a project schedule

- **Slack time** = available time – real time
  = latest start time – earliest start time

CPM Bar Chart

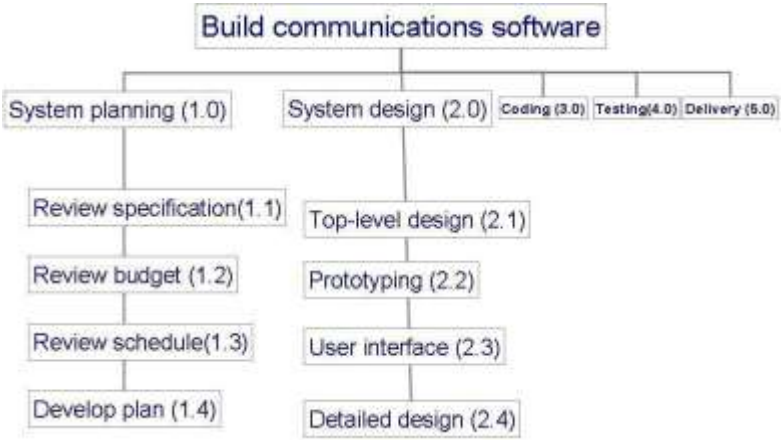
- Including information about the early and late start dates
- Asterisks indicate the critical path



Tools to Track Progress

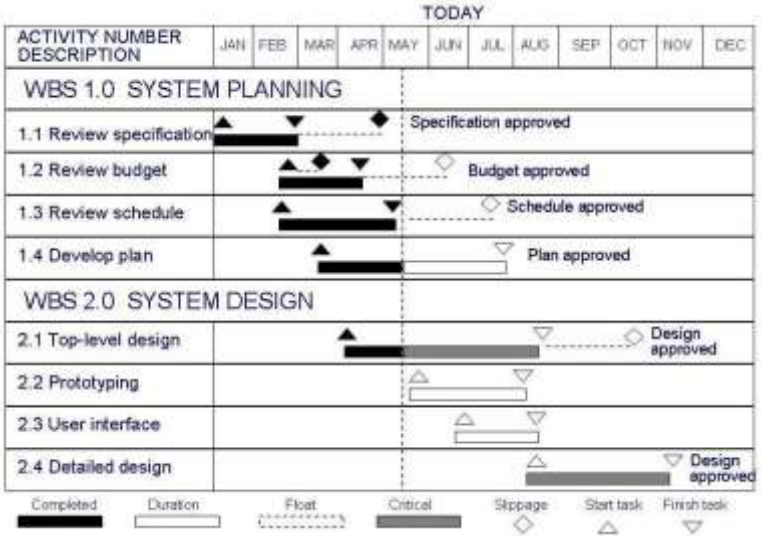
- Example:

  to track progress of building a communication software



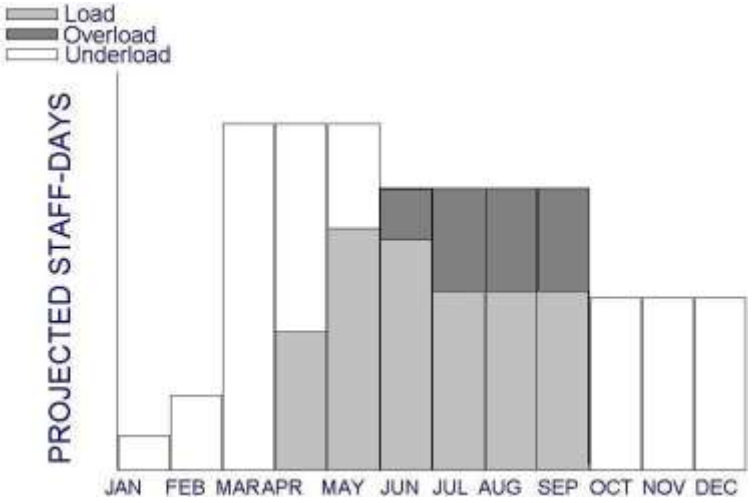
Tools to Track Progress: Gantt Chart

- Activities shown in parallel
  - helps understand which activities can be performed concurrently



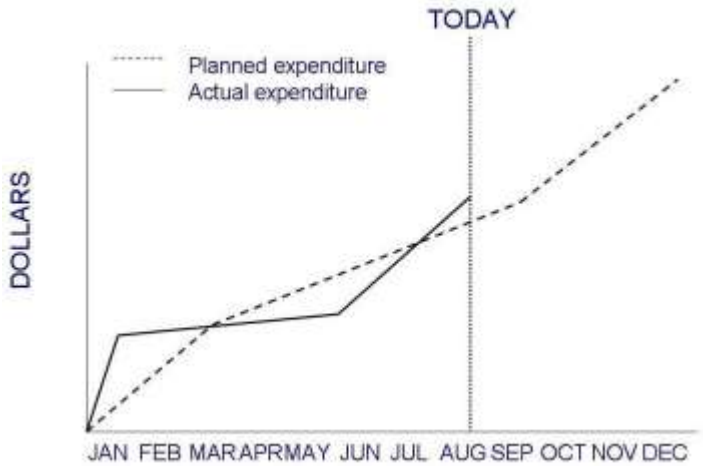
Tools to Track Progress: Resource Histogram

- Shows people assigned to the project and those needed for each stage of development



Tools to Track Progress: Expenditures Tracking

- An example of how expenditures can be monitored



Staffing

- Key activities requiring personnel
  - requirements analysis
  - system design
  - program design
  - program implementation
  - testing
  - training
  - maintenance
  - quality assurance
- There is great advantage in assigning different responsibilities to different people.

Choosing Personnel

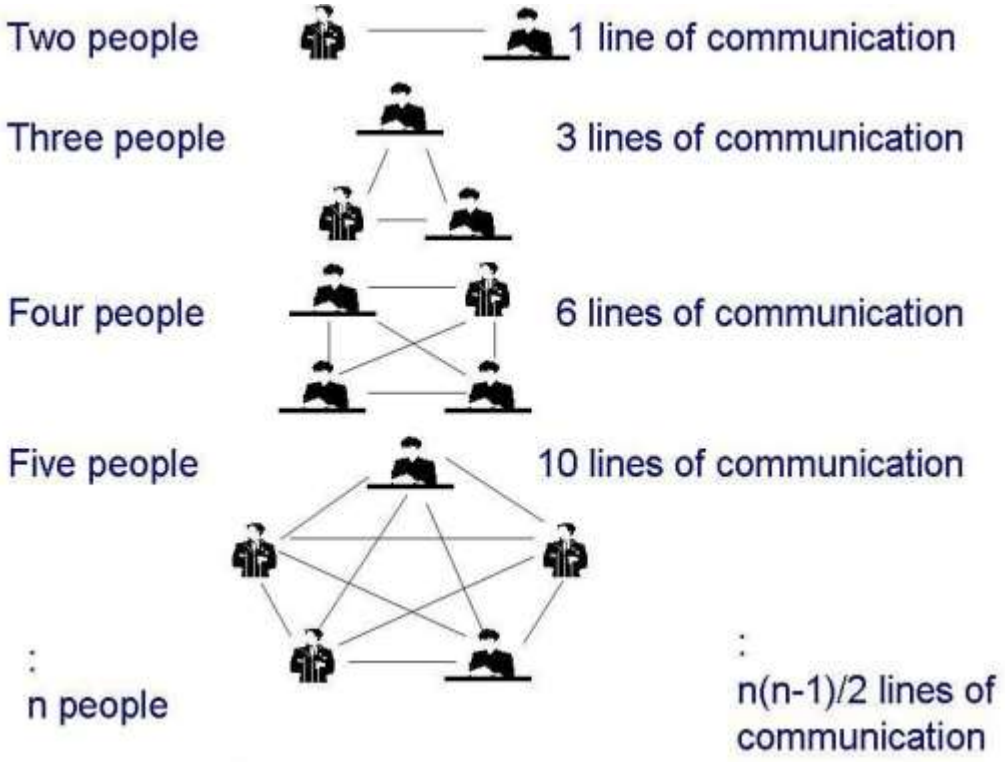
- **Ability** to **perform work.**
- **Interest** in work.
- **Experience** with
  - similar applications
  - similar tools, languages, or techniques
  - similar development environments
- **Training.**
- Ability to **communicate** with others.
- Ability to **share responsibility.**
- **Management skills.**

Communication

- A project's **progress** is affected by
  - **degree of communication.**
  - **ability of individuals to communicate their ideas.**
- S/w **failures** can result from **breakdown in communication and understanding**.

Communication

- **Line of communication** can grow quickly
- If there is ***n*** worker in project, then there are ***n(n-1)/2*** pairs of communication



Make Meeting Enhance Project Progress

- **Common complains about meeting**
  - the purpose is unclear
  - the attendees are unprepared
  - essential people are late or absent
  - the conversation veers away from its purpose
  - participants do not discuss, instead argue
  - decisions are never enacted afterward
- **Ways to ensure a productive meeting**
  - clearly decide who should be in the meeting
  - develop an agenda
  - have someone who tracks the discussion
  - have someone who ensures follow-up actions

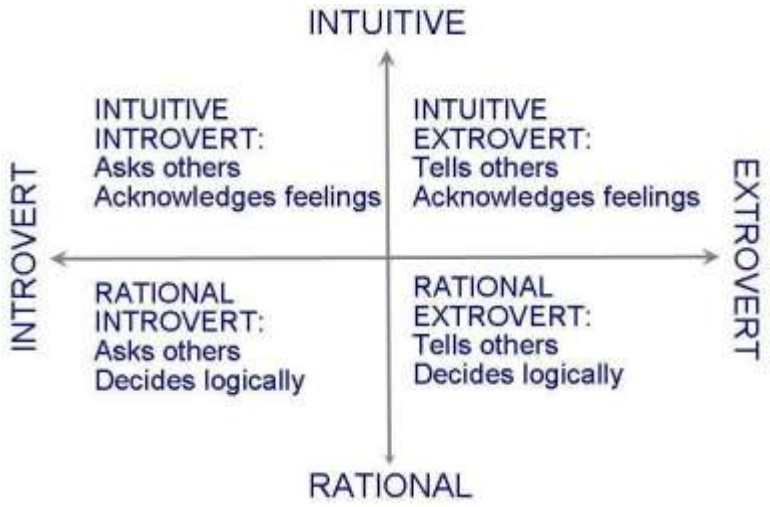
Work Styles

1) **Extroverts:** tell their thoughts.
2) **Introverts:** ask for suggestions.
3) **Intuitives:** base decisions on feelings.
4) **Rationals:** base decisions on facts, options.

A strong team listens to all these styles.

Work Styles

- Horizontal axis: communication styles
- Vertical axis: decision styles



Work Styles

- Work styles determine communication styles
- Understanding workstyles
  - help to be flexible
  - give information based on other's priorities
- Impacts interaction among customers, developers and users

Project Organization

- Depends on

  1) backgrounds and **work styles** of team members.

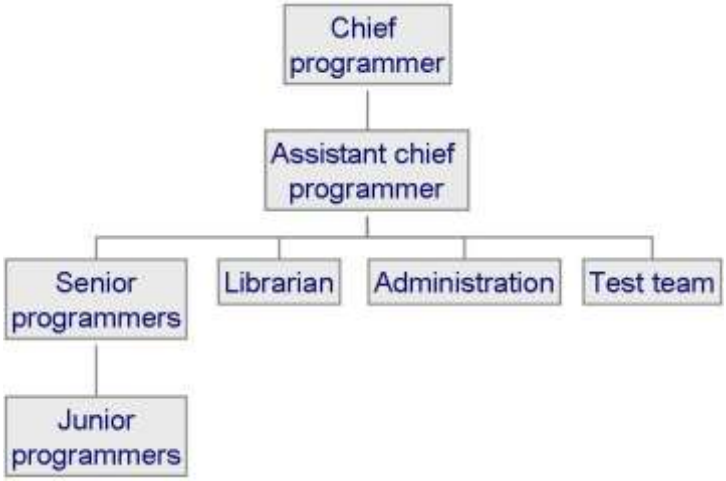
  2) number of people on team.

  3) management styles of customers and developers.

- Examples:
  - ***Chief programmer team*:** one person totally responsible for a system's design and development
  - ***Egoless approach:*** *hold everyone equally responsible*

Project Organization: Chief Programmer Team

- Each team member must communicate often with chief, but not necessarily with other team members



Project Organization

- **Characteristics of projects** and the **suggested organizational structure** to address them

Highly structured	Loosely structured
High certainty	Uncertainty
Repetition	New techniques or technology
Large projects	Small projects

Structure vs. Creativity

- Experiment by **Sally Phillip** examining two groups **building a hotel**
  - **structured team: clearly defined responsibilities.**
  - **unstructured team: no directions.**

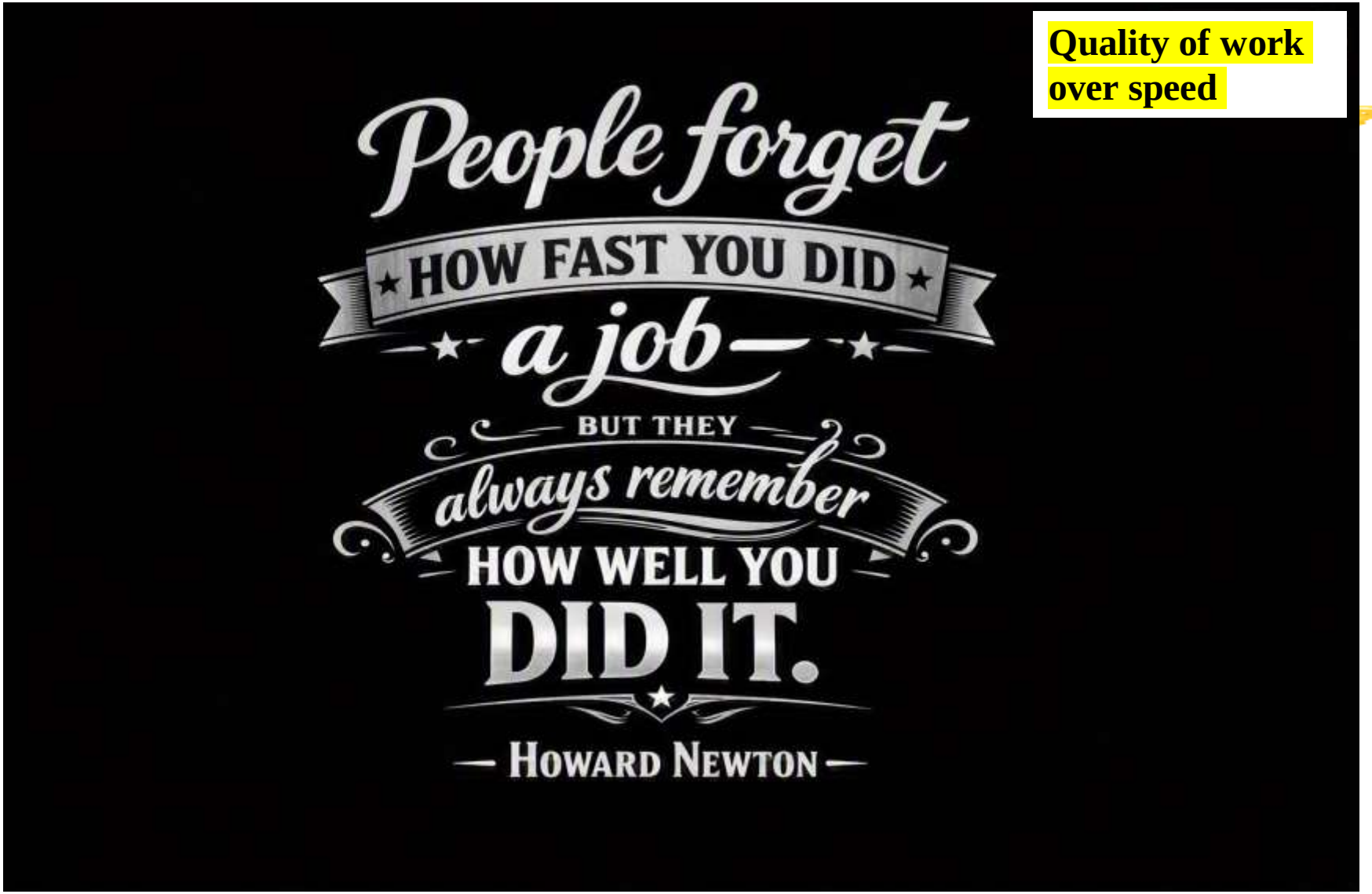
- The results are always the same
  - Structured teams *finish* a functional Days Inn.
  - Unstructured teams build a **creative, multistoried** Taj Mahal and never complete.

- Good project management means finding a **balance between structure and creativity.**

Quality Planning



Quality of work over speed



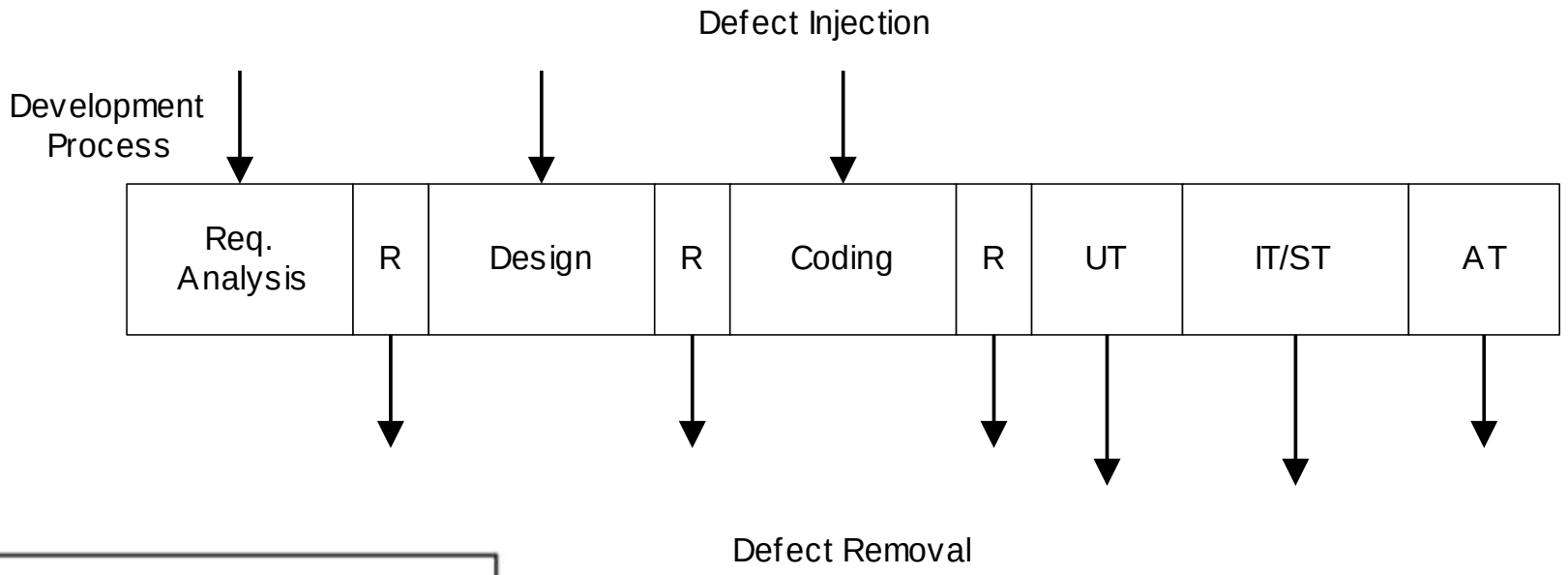
Quality Planning

- Delivering high quality is a basic goal
- Quality can be defined in many ways(secure, fast, easy, do not crash, works correctly)
- Current industry standard - *delivered defect density* (e.g. #defects/KLOC)
- Defect - something that causes software to behave in an inconsistent/incorrect manner
- Aim of a project - deliver software with low delivered defect density

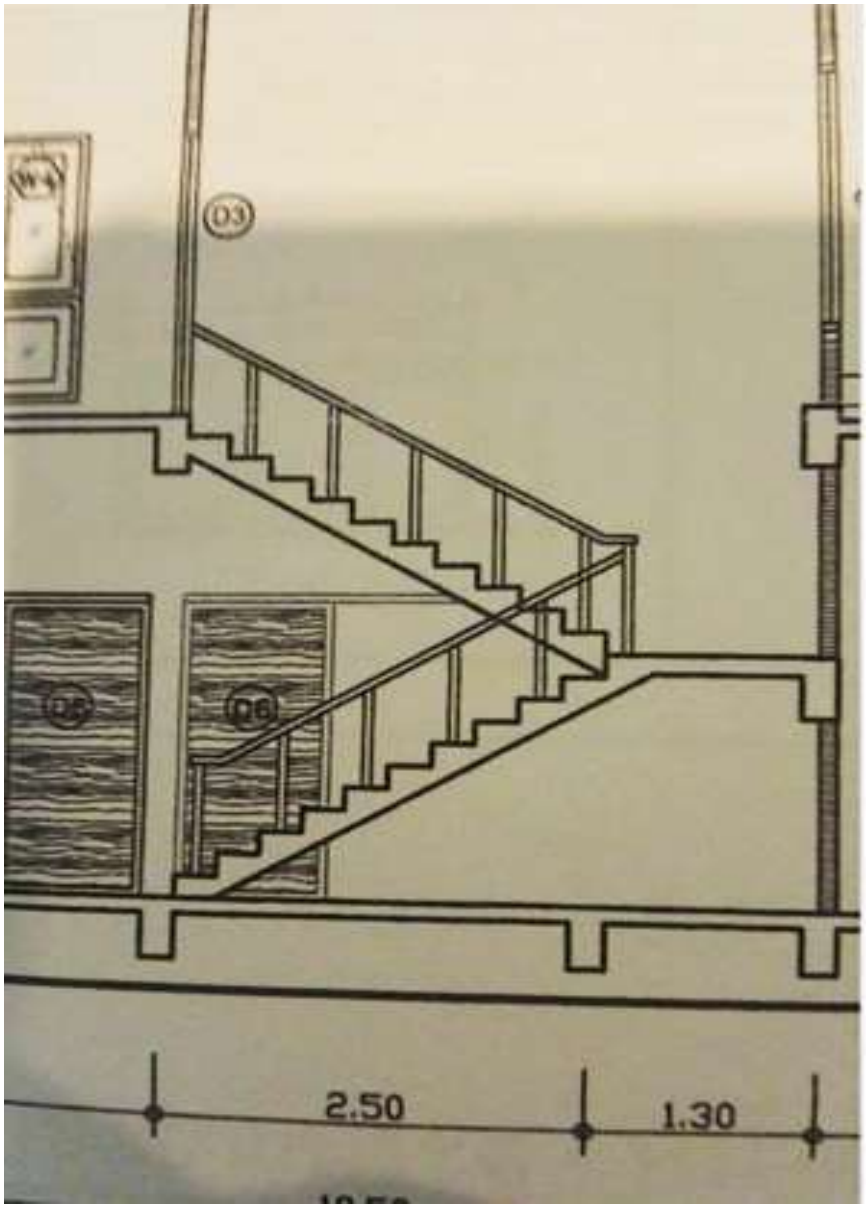
Defect Injection and Removal

- Software development is a highly people-oriented activity and hence it is error-prone.
- Defects are injected at any stage
- As quality goal is low delivered defect density, these defects have to be removed
- Done primarily by quality control (QC) activities of reviews and testing

Defect Injection and Removal



R – Removal
UT – Unit Testing
IT – Integration Testing
ST – System Testing
AT – Acceptance Testing


03
05
06
2.50
1.30



Approaches to Quality Management

- Ad hoc - some testing, some reviews done as and when needed
- Procedural - defined procedures are followed in a project
- Quantitative - defect data analysis done to manage the quality process

Procedural Approach

- A quality plan defines what QC tasks will be undertaken and when
- Main QC tasks - reviews and testing
- Guidelines and procedures for reviews and testing are provided
- During project execution, adherence to the plan and procedures ensured

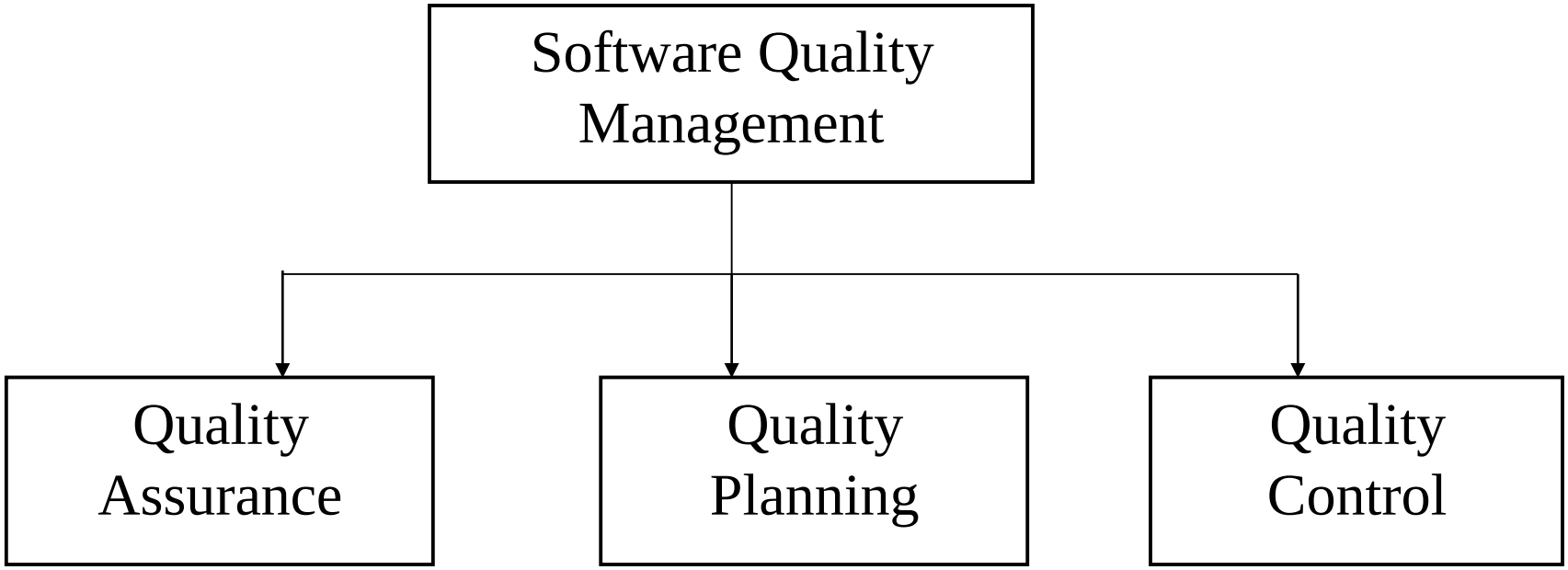
Quantitative Approach

- Goes beyond asking "has the procedure been executed"
- Analyzes defect data to make judgements about quality
- Past data is very important
- Key parameters - defect injection and removal rates, defect removal efficiency (DRE)

Software Quality Management

- Software Quality Management (SQM) refers to the complete process that ensures a software product is develop as per national and international standards like ANSI, IEEE and ISO.
- Need of Software Quality Management:
  - Delivering high-quality products on time .
  - Increases stakeholder faith on product & company.
  - High-quality products always ensure customer satisfaction.

Activities of Software Quality Management:



How to Achieve Software Quality?

Quality Assurance:

- It assure that system meets specified requirements and customer expectations.
- It defines standards and methodologies for successful development process.
- It assure Correctness, Efficiency, Flexibility, Maintainability, Portability, Usability etc.

Quality Control:

- It focuses on to achieve & fulfill quality parameters or quality goals as per customer requirements.
- It focus on deliver product on time with accurate cost.

Quality Planning:

- Select applicable procedures and standards for a particular project and modify as required to develop a quality plan.

Quality Plan

- The quality plan drives the quality activities in the project
- Level of plan depends on models available
- Must define QC tasks that have to be performed in the project
- Can specify defect levels for each QC tasks (if models and data available)

- The quality plan ensures quality is planned from the beginning, checked throughout the project, and measured when enough data is available.

Risk Management



Project Risks



Risk Management

- Any project can fail - reasons can be technical, managerial, etc.
- Project management aims to tackle the project management aspect
- Engineering life cycles aim to tackle the engineering issues
- A project may fail due to unforeseen events - risk management aims to tackle this

Risk Management

- **Risk:** any condition or event whose occurrence is not certain but which can cause the project to fail.
- **Aim of risk management**: minimize the effect of risks on a project
- Risk management has two basic aspects
  - Risk assessment
  - Risk control

Risk Assessment

- To identify possible risks to a project, i.e. to those events that might occur and which might cause the project to fail
- No "algorithm" possible, done by "what ifs", checklists, past experience
- Can have a list of "top 10" risks that projects have seen in past

Top Risk Examples

- Shortage of technically trained manpower
- Too many requirement changes
- Unclear requirements
- Not meeting performance requirements
- Unrealistic schedules
- Insufficient business knowledge
- Working on new technology

Risk Prioritization

- The number of risks might be large
- Must prioritize them to focus attention on the "high risk" areas
- For prioritization, impact of each risk must be understood.(how bad would it be if this actually happen)
- In addition, probability of the risk occurring should also be understood.(how likely is this to happen)

Risk Prioritization ...

- Risk exposure (RE) = probability of risk occurring * risk impact
- RE is the expected value of loss for a risk
- Prioritization can be done based on risk exposure value
- Plans can be made to handle high RE risks

- There's a **30% chance** your teammate quits the project (probability = 0.3)
- If they DID quit, it would cost you about **10 hours** of extra work to fix it (impact = 10)

Then: **Risk Exposure = 0.3 × 10 = 3**

A Simple approach to Risk Prioritization

- Classify risk occurrence probabilities as: Low, Medium, High
- Classify risk impact as: Low, Medium, High
- Identify those that are HH, or HM/MH
- Focus on these for risk mitigation
- Will work for most small and medium sized projects

Risk Control

- Can the risk be avoided?
  - E.g. if new hardware is a risk, it can be avoided by working with proven hardware
- For others, risk mitigation steps need to be planned and executed
  - Actions taken in the project such that if the risk materializes, its impact is minimal
  - Involves extra cost

Risk Management Paradigm

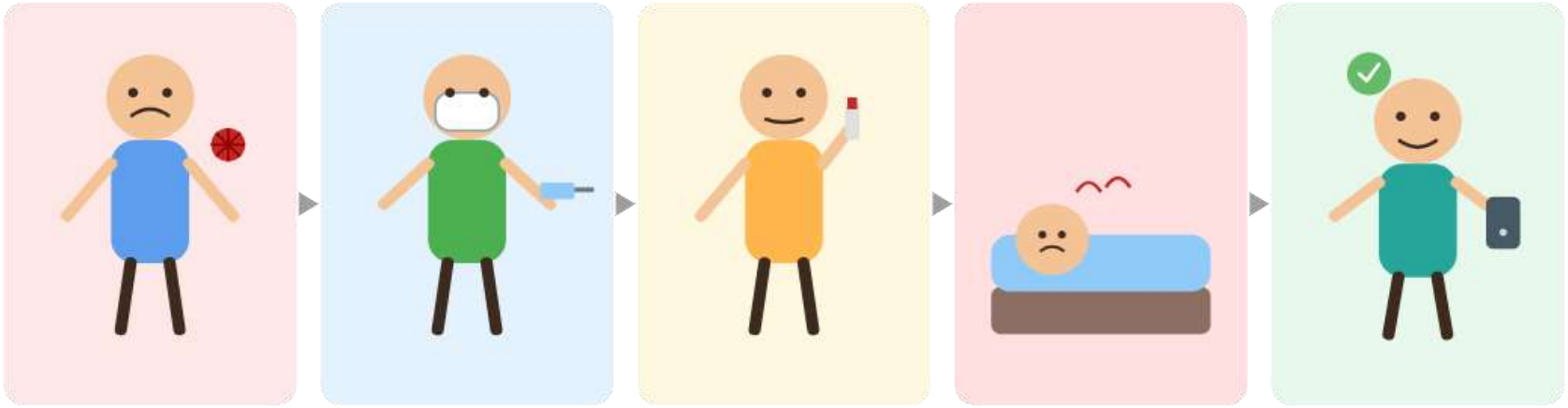




RMMM Plan

Risk Mitigation, Monitoring and Management Plan

A Real Life Example



About RMMM

- A risk management technique is usually seen in the software Project plan.
- This can be divided into **Risk Mitigation, Monitoring and Management Plan** (RMMM).
- The project manager typically uses this RMMM plan as part of the overall project plan.
- Risk is documented with the help of a Risk Information Sheet (RIS).
- RIS maintain Risk ID, Date, Probability, Impact, Description, Avoidance, Monitoring, Management plan & Current status.
- This RIS is controlled by using a database system for easier management of information i.e creation, priority ordering, searching and other analysis.

Building a Risk Table

Risk	Probability	Impact	RMMM
			Risk Mitigation Monitoring & Management

Risk Mitigation

Risk Mitigation is a technique for avoiding risks (Risk Avoidance).
It is proactive approach. Apply before risk have generate.

➢ **Steps for mitigating the risks as follows:**

1. Communicate with concerned staff to find off probable risk.
2. Removing all causes that are the reason for risk creation.
3. Develop a policy in an organization which will help to continue the project.
4. Controlling the corresponding documents from time to time.
5. Conducting timely reviews to speed up the work.

Risk Monitoring

- Risk monitoring is an activity used to track a project's progress.
- Performed by Project Manager.

➢ **Objectives of Risk Monitoring Process**:

1. To check if predicted risks occur or not.
2. To ensure proper application of risk avoidance are apply or not.
3. To gather information for future risk assessments.
4. To determine which risks generate which problems throughout the project.

If (RISK is avoidable)
{
- **Risk Mitigation. //(1)**
- **Risk Monitoring. //(2)**

}
else //(i.e. RISK is unavoidable)
{
- **Risk Management. //(3)**

}

Risk Management

- It is reactive approach, Apply after risk have generate.
- It assumes that the mitigation activity failed and the risk generate in reality.
- This task is done by Project manager, They will solve generated risk.
- If the project manager effectively uses project mitigation to remove risks successfully then it is easier to manage the risks.

Example:
Consider a scenario that many people leaving the organization but

✓ If sufficient additional staff is available
✓ If current development activity knows to everybody in the team
✓ If systematic documentation available

Then any new employee easily understand & start current development activity.

RMMM Plan Example

➢ Risk generated : Late Delivery of Project

1. **Mitigation (Avoid Risk):**
- Before development apply some precautionary measures.
- Developer already knew Project will be complete in 20 days. But he told customer Project will complete in 30 days.

2. **Monitoring:**
- To develop project schedule. Mentioned start & end date of Project. Within 20 to 30 days.

3. **Management:**
- Project not completed within deadline then negotiate with customers.
- Ask some extra time or give some additional features etc.